

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Scientific methodologies & sampling design - Set 1

Mark scheme

AQA spec 3.7.1; Research methods; textbook pp. 390-393

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• There is an environmental gradient (distance from the hedge / changing conditions) so samples must be taken at regular intervals along it to detect the change • Random sampling might leave gaps / miss the pattern; systematic sampling avoids bias while covering the gradient	2
01.2	• Sample at close intervals (eg every 1 m) first and see at what spacing the variations are still detected without unnecessary data; wider intervals (eg 4 m) may miss peaks and troughs • Take increasing numbers of samples at one site and calculate the running mean; the number needed is where the mean stops fluctuating / stabilises • Calculate standard deviation to assess scatter: more variable data need more samples • Test the method (extraction technique, timing) for reliability before the main study	4
01.3	• Remove surface vegetation; pour an irritant solution (dilute mustard / detergent in water) onto the area until saturated (and around it), keep adding as it infiltrates • Collect worms that come to the surface for a standardised time; count / identify / weigh • Limitations: solution may not percolate impermeable soils or reach deep worms; worms may move sideways or downwards; some may die or not respond; slow-moving worms missed	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393; AQA spec 3.7.1.	
02.1	• About 8 nights • The running mean fluctuates widely with few samples but stabilises at 21-22 from 8 nights onwards; extra samples do not change it significantly	2

Question	Marking guidance	Marks
02.2	• Moth activity varies with temperature, wind, rain and season (adults only fly for part of the year) so catches vary independently of population size • Sampling at different conditions / times would make results not comparable • Timing between samples should be chosen to detect the temporal variation of interest (eg weekly through the season)	3
02.3	• Accuracy: how close a recorded result is to the real value • Precision: the smallest interval between possible recorded values (eg a ruler marked in mm is more precise than one in cm)	2
02.4	• A null hypothesis (eg no difference / no correlation) can be tested statistically and accepted or rejected at a stated confidence level • Standardised techniques ensure results collected at different times, places or by different people are comparable • Other variables are controlled or monitored so their effect can be assessed; allows the study to be repeated / checked (reliability)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393; AQA spec 3.7.1.	
03.1	• Trays of the same soil (texture, compaction, organic matter, depth) with different percentage vegetation cover (eg 0, 25, 50, 75, 100% grass), replicated • Same slope / gradient; simulated rain from a standard height at a standard rate and drop size (eg watering can / sprinkler) for a standard time • Collect the soil splashed / washed off (eg onto a sheet or into a trough at the base), dry and weigh it • Repeat each treatment several times and calculate means (and standard deviation) • Plot soil loss against cover; statistical test (Spearman's rank / t-test) for significance	5
03.2	<i>Any 3 from:</i> • Rainfall intensity / raindrop size / height • Soil texture, compaction, organic matter, depth • Permeability of the material beneath • Gradient • Root binding	3
03.3	• All variables except the independent variable can be standardised in the laboratory; in the field uncontrollable variables (weather) must be measured instead • Laboratory conditions may not represent natural conditions so results may not apply directly to real ecosystems	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393, 419-420; AQA spec 3.7.1, 3.7.2.2.	
04.1	• B	1
04.2	• Number all possible sampling sites (eg on a grid map) • Select the sites using a random number generator / table; (regular shape: grid coordinates from random numbers)	2
04.3	• Sites may be chosen for convenience or because they appear to support (or dismiss) the hypothesis • So the sample does not represent the whole area and conclusions are invalid	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 391-393; AQA spec 3.7.1.	
05.1	The readings are precise but may not be accurate	1
05.2	- A result that differs from other results with which it was expected to be similar - Repeat the study / measurement: if it recurs it is a 'real' result caused by another variable; if not it was an error and can be excluded	2
05.3	- To test the reliability of the method / equipment - To decide the sample size, number of samples, spacing or timing needed to detect the variation without collecting unnecessary data	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-391; AQA spec 3.7.1.	
06.1	- About 40-50 traps - Because the running mean stops changing / fluctuating with fewer than this the mean is unreliable	2
06.2	- Widely scattered results (large standard deviation) mean a single sample may not be representative - More samples are needed so that the effect of variability is averaged out and the mean is representative / significance can be assessed	2
06.3	<i>Any 3 from:</i> - Same type and size of pitfall trap, set for the same length of time - Traps emptied at the same time of day / same weather conditions in both habitats - Same number of traps and same method of positioning (random or systematic) in each habitat - Other variables that could affect results (eg rainfall, temperature) monitored or controlled	3
06.4	- Flying insect activity changes with weather over minutes to weeks (temperature, wind) and diurnally - Samples on different occasions are needed to produce a mean that represents the typical value or to identify trends - The interval between samples should match the rate at which the factor changes, found from a preliminary study	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 391-393; AQA spec 3.7.1.	